Shanmugam Subramanian

Mechanical Engineer II | Lightweight Structures & Aerodynamics | Hyderabad, India

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SUMMARY

Mechanical engineer with 4 years of NPD and NPI experience in lightweight structural design, aerodynamics and rapid prototyping for antenna and radar hardware. Cut 60 g per unit with topology-optimized 3D-printed parts, ran Ansys Fluent CFD on 135+ radome profiles (12–18% wind load reduction), and designed parametric drone propeller blades from NACA/Eppler airfoils. Experienced in vibration analysis and FEA.

EXPERIENCE

JMA Wireless — Hyderabad, India
Mechanical Engineer II May 2023 – Present

- Applied **topology optimization** to lightweight FDM 3D-printed prototypes, **cutting 60 g per unit** while meeting stiffness requirements.
- Optimized radome aerodynamics with **Ansys Fluent CFD** (2D/3D): design of experiments on 135+ profiles achieved **12–18% wind load reduction**. Documented workflows for team adoption & working toward patent filing.
- Performed **harmonic & random vibration analyses** with 80% first-pass correlation to physical tests (IEC 60068-2-6/64).
- Executed **static structural analysis** with FOS targets of 1.5–3 (metals) & 2–5 (plastics).
- Built Excel-based **hand-calculation tools** to estimate drag force from geometry, crimping force for pneumatic cylinder sizing, and snap-joint behavior in plastic part design.
- Performed **IEC 60068-2-31 drop tests** for mechanical qualification of antenna products.
- Owned end-to-end **NPD & NPI** for eight 5G Macro & Distributed Antenna Systems products for the US & Canada markets.
- Released RF PCBs, sheet metal, injection-molded, die-cast, CNC, EDM, & FDM parts with a **90% FAI pass rate**.
- Automation & process-improvement by developing RF PCB sub-assembly fixtures, rivet press-pull tools, & alignment jigs, achieving **40% rework reduction, 75% faster riveting, and cutting up to 20–50 mins per antenna** in the production line.
- Managed **ECRs, FAIs, & SVPRs** using Salesforce Propel **PLM** & SolidWorks **PDM** in coordination with the global team.
- Supported **contract manufacturing** setup in Chennai through BOM costing, tooling transfer, & operator training.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Design Engineer (R&D) Jul 2022 – Apr 2023

- Developed two automotive radar antenna chassis and radomes with **IP68 (IEC 60529)** ratings for ADAS applications.
- Enhanced chassis thermal properties by refining fin geometry, reducing **internal temperature by 6°C**.
- Lowered radome **cost by 8%** through material optimization without degrading dielectric performance.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Project Intern Mar 2022 – Jun 2022

- Updated automotive radar housing designs based on lead engineer feedback & prepared GD&T-compliant drawings.
- Implemented **snap-fit** radome design, reducing the radar BOM by two components and saving $0.3 per unit.

EDUCATION

Bachelor of Engineering — Mechanical Engineering GPA: 8.21 / 10
Alagappa Chettiar College of Engineering and Technology (Anna University) 2018 – 2022

PROJECTS AND COMPETITIONS

Drone Propeller Blade Design Personal project
Designed parametric propeller blades in multiple diameter/pitch variants from NACA/Eppler airfoil profiles.

Venturi Fan Blade Mechanism Patent in progress
CFD-validated ducted fan blade concept for secondary jet flow augmentation; patent filing in progress.

FDM Subsonic Wind Tunnel Personal project
Designed and 3D-printed a functional subsonic wind tunnel for aerodynamic testing of 1:16 scale die-cast cars.

Non-conventional energy sources - Paper presented— Dec 2019 VCET, Madurai
Designed a roof-mounted micro-turbine array for vehicle-based wind energy harvesting, generating (5–8 W) at highway speeds.

SAE Bicycle Design Competition — 2020 - 2021 SAE INDIA
Developed a planetary gear-based bicycle drivetrain achieving 15% higher torque than a typical bicycle.

SKILLS & SOFTWARE

Design: SolidWorks, AutoCAD, Fusion 360, Inventor, Onshape, GD&T, Tolerance stackup, 3D Printing, **3D Printing (FDM)**, lightweight design, airfoil (NACA/Eppler) blade design
Simulation: Ansys Mechanical & Fluent, Finite Element Analysis, SolidWorks Simulation & Motion Analysis, topology optimization, CFD, hand-calculation tools (Excel)
Programming & Embedded systems: Python, MATLAB Simulink & Simscape, C, C++, Arduino
Other Tools: Excel, PowerPoint, PLM (Salesforce Propel), SolidWorks PDM
Standards & Testing: IEC 60068-2-6/-64/-31, IEC 60529 (IP68)